

ANSWER KEY

GATE EC — GATE EC 2026 Trend-Based Mock Paper
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Q.No.	Type	Answer	Marks
1	2MCQ	B	2
2	2NAT	4	2
3	2MCQ	B	2
4	2MCQ	6	2
5	2NAT	3	2
6	2MCQ	B	2
7	3MCQ	3	3
8	2MCQ	A	2
9	3MCQ	B	3
10	2MCQ	C	2
11	2NAT	would have become	2
12	2NAT	A	2
13	2NAT	B	2
14	1NAT	B	1
15	1NAT	B	1
16	2MCQ	B	2
17	2MCQ	C	2
18	2MCQ	A	2
19	2MSQ	A,B	2
20	2MCQ	6.4	2
21	2MCQ	B	2
22	2NAT	0.5	2
23	2MCQ	C	2
24	1MCQ	B	1
25	1MCQ	B	1
26	1MCQ	B	1
27	1MCQ	B	1
28	1MCQ	B	1
29	1NAT	C	1
30	1MCQ	B	1
31	1MCQ	C	1
32	1MCQ	A	1
33	1NAT	A	1
34	1MCQ	B	1
35	1MCQ	A	1
36	1MCQ	A	1

37	1MCQ	B	1
38	1MCQ	A	1
39	1MCQ	A	1
40	1MCQ	B	1
41	1NAT	A	1
42	1MCQ	A	1
43	1NAT	A	1
44	1MCQ	A	1
45	1NAT	C	1
46	1MCQ	A	1
47	1NAT	D	1
48	1MCQ	B	1
49	1MCQ	B	1
50	2NAT	2	2
51	2NAT	0.48	2
52	2MCQ	A	2
53	2MCQ	B	2
54	2MCQ	D	2
55	1MCQ	B	1
56	2MSQ	A,B	2
57	1MCQ	A	1
58	2NAT	2	2
59	2NAT	3	2
60	2NAT	0.5	2
61	1MCQ	B	1
62	2MSQ	A, D	2
63	1MCQ	Q,S,P,R	1
64	2MCQ	A	2
65	2MSQ	B, D	2

Q1. General Aptitude | 2 marks | 2MCQ
Sum of first 100 natural numbers:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q10

Q2. General Aptitude | 2 marks | 2NAT
The 12 musical notes are given as $\sharp, \sharp, \flat, \sharp, \flat, \sharp, \flat$ and $\flat$. Frequency of each note is 122 times the frequency of ...

Answer: 4

Musical notes frequency ratio $2^{(1/12)}$. 12 notes per octave. The pattern repeats every 12 notes.

Source: Adapted from GATE 2025 EC Session 1 Q3

Q3. General Aptitude | 2 marks | 2MCQ
If train 100m crosses pole in 10s, speed:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q10

Q4. General Aptitude | 2 marks | 2MCQ
For a real number $x > 1$, $\frac{1}{x} + \log_2 x + \log_3 x + \log_4 x = 1$ The value of x is

Answer: 6

$\log_2 x + \log_3 x + \log_4 x = 1 - \frac{1}{x}$ Using change of base: $\ln; \frac{1}{x} = \frac{\ln x}{\ln x}$ Solving gives $x = 6$

Source: Adapted from GATE 2024 EC Session 7 Q4

Q5. General Aptitude | 2 marks | 2NAT
Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify th...

Answer: 3

Analogous groups: decreasing order of intensity. Mapping words between groups based on intensity levels.

Source: Adapted from GATE 2025 EC Session 1 Q1

Q6. General Aptitude | 2 marks | 2MCQ
Choose synonym of 'trivial':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q10

Q7. General Aptitude | 3 marks | 3MCQ
The greatest prime factor of $(3199 - 3196)$ is

Answer: 3

$3199 - 3196 = 3$, and 3 is prime, so its greatest prime factor is 3.

Source: Adapted from GATE 2024 EC Session 7 Q5

Q8. General Aptitude | 2 marks | 2MCQ
The following figures show three curves generated using an iterative algorithm. The total length of the curve generated ...

Answer: A

Each iteration multiplies the number of segments by 3 while each segment's length is reduced by a factor of $1/3$, so the total length remains constant (1). Hence the correct choice is option A.

Source: Adapted from GATE 2025 EC Session 1 Q4

Q9. General Aptitude | 3 marks | 3MCQ

Which one of the following plots represents $f(x) = -|x|$, where x is a non-zero real number? Note: The figures shown are rep...

Answer: B

$f(x) = -|x|$ is a V-shaped graph opening downward. Option B shows this shape.

Source: Adapted from GATE 2025 EC Session 1 Q5

Q10. General Aptitude | 2 marks | 2MCQ

If ' $\sim$ ' denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to [bored...

Answer: C

Charm < enamor < bewitch (increasing intensity). Bored < intrigued < fascinated. 'Intrigued' fits the middle intensity.

Source: Adapted from GATE 2024 EC Session 7 Q1

Q11. General Aptitude | 2 marks | 2NAT

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above ...

Answer: would have become

The clause 'Had I learnt acting as a child' is a past unreal conditional (inverted 'If I had learnt...'). The result clause requires the perfect conditional form, 'would have become', to indicate a hypothetical outcome in the past.

Source: Adapted from GATE 2025 EC Session 1 Q2

Q12. General Aptitude | 2 marks | 2NAT

Choose opposite of 'benevolent':

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q10

Q13. Engineering Mathematics | 2 marks | 2NAT

$i^4 = ?$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q1

Q14. Engineering Mathematics | 1 marks | 1NAT

$P(A \cap B) = ?$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q1

Q15. Engineering Mathematics | 1 marks | 1NAT

Integral of $1/x$ dx:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q1

Q16. Network, Signals & Systems | 2 marks | 2MCQ

A full adder and an XOR gate are used to design a digital circuit with inputs X, Y, and Z, and output F, as shown below....

Answer: B

Full adder with XOR gate design. Logic analysis shows output F matches option B for given inputs.

Source: Adapted from GATE 2025 EC Session 1 Q24

Q17. Network, Signals & Systems | 2 marks | 2MCQ

Consider the unity-negative-feedback system shown in Figure (i) below, where gain 0 . The root locus of this system is sh...

Answer: C

Unity feedback system with gain $K > 0$. Root locus analysis shows correct path matching option C.

Source: Adapted from GATE 2025 EC Session 1 Q26

Q18. Network, Signals & Systems | 2 marks | 2MCQ

The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the p...

Answer: A

Electron concentration profile in doped semiconductor at equilibrium. Analysis shows option A is correct.

Source: Adapted from GATE 2024 EC Session 1 Q26

Q19. Network, Signals & Systems | 2 marks | 2MSQ

As shown in the circuit, the initial voltage across the capacitor is 10 V , with the switch being open. The switch is the...

Answer: A,B

RC circuit with initial capacitor voltage 10V . Switch closes at $t=0$. Analysis shows currents/voltages match options A and B.

Source: Adapted from GATE 2024 EC Session 1 Q35

Q20. Network, Signals & Systems | 2 marks | 2MCQ

A $50\text{ }\Omega$ lossless transmission line is terminated with a load Z_L of $(50 - j75)\text{ }\Omega$. If the average incident power on the line is...

Answer: 6.4

The reflection coefficient $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{50 - j75 - 50}{50 - j75 + 50} = \frac{-j75}{100 - j75} = \frac{75}{125} = 0.6$, so $|\Gamma|^2 = 0.36$. The transmitted power fraction is $1 - |\Gamma|^2 = 0.64$. With an incident average power of 10 mW , the power delivered to the load is $10\text{ mW} \times 0.64 = 6.4\text{ mW}$ (rounded to one decimal place).

Source: Adapted from GATE 2025 EC Session 1 Q64

Q21. Network, Signals & Systems | 2 marks | 2MCQ

A stick of length one meter is broken at two locations at distances of 1 and 2 from the origin (0), as shown in the figu...

Answer: B

Stick broken at two locations (1 and 2). Probability analysis for forming triangle gives answer matching option B.

Source: Adapted from GATE 2025 EC Session 1 Q7

Q22. Network, Signals & Systems | 2 marks | 2NAT

The diode in the circuit shown below is ideal. The input voltage (in Volts) is given by $v(t) = 10 \sin 100t$, where time is in s...

Answer: 0.5

Ideal diode with sinusoidal input $v(t) = 10 \sin(100t)$. During positive half-cycle, diode conducts. Average output voltage $= 0.5 \times 10 = 0.5\text{ V}$.

Source: Adapted from GATE 2025 EC Session 1 Q61

Q23. Network, Signals & Systems | 2 marks | 2MCQ

The intrinsic carrier concentration of a semiconductor is $2.5 \times 10^{16} / \text{m}^3$ at 300 K . If the electron and hole mobilities a...

Answer: C

Intrinsic carrier concentration $n_i = 2.5 \times 10^{16} / \text{m}^3$. Using conductivity formula $\sigma = q(n\mu_n + p\mu_p)$, calculate conductivity.

Source: Adapted from GATE 2025 EC Session 1 Q46

Q24. Electronic Devices | 1 marks | 1MCQ

SCR has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q9

Q25. Electronic Devices | 1 marks | 1MCQ
n-type majority carriers:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q3

Q26. Electronic Devices | 1 marks | 1MCQ
MOSFET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q8

Q27. Electronic Devices | 1 marks | 1MCQ
Zener diode used for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q9

Q28. Analog Circuits | 1 marks | 1MCQ
Common emitter has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q4

Q29. Analog Circuits | 1 marks | 1MCQ
Ideal op-amp open-loop gain:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 EC Session 1 Q2

Q30. Analog Circuits | 1 marks | 1MCQ
FET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q3

Q31. Analog Circuits | 1 marks | 1MCQ
RC oscillator frequency:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 EC Session 1 Q3

Q32. Analog Circuits | 1 marks | 1MCQ
RC oscillator frequency:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q9

Q33. Digital Circuits | 1 marks | 1NAT
ROM is:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q4

Q34. Digital Circuits | 1 marks | 1MCQ
D flip-flop follows input after:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q5

Q35. Digital Circuits | 1 marks | 1MCQ
4-bit binary counter counts:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q4

Q36. Digital Circuits | 1 marks | 1MCQ
TTL stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q3

Q37. Control Systems | 1 marks | 1MCQ
Transfer function is defined:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q5

Q38. Control Systems | 1 marks | 1MCQ
Nyquist criterion uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q6

Q39. Control Systems | 1 marks | 1MCQ
Routh-Hurwitz determines:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q7

Q40. Control Systems | 1 marks | 1MCQ
Lead compensator:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q6

Q41. Communication Systems | 1 marks | 1NAT
ASK modulation varies:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q5

Q42. Communication Systems | 1 marks | 1MCQ
FM has:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q5

Q43. Communication Systems | 1 marks | 1NAT
PCM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q4

Q44. Communication Systems | 1 marks | 1MCQ
AM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q6

Q45. Electromagnetics | 1 marks | 1NAT
Maxwell has how many equations:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 EC Session 1 Q8

Q46. Electromagnetics | 1 marks | 1MCQ
Characteristic impedance Z_0 =:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q7

Q47. Electromagnetics | 1 marks | 1NAT
Skin depth increases with:

Answer: D

Correct Answer: D

Source: Adapted from GATE 2023 EC Session 1 Q6

Q48. Electromagnetics | 1 marks | 1MCQ
Waveguide supports:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q7

Q49. Control Systems | 1 marks | 1MCQ
Transfer function is defined:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q5

Q50. Network, Signals & Systems | 2 marks | 2NAT
 $s(s+1)(s+2)R(s) + E(s)G(s)Y(s) - R(s)$, $Y(s)$, and $E(s)$ are the Laplace transforms of $r(t)$, $y(t)$, and $e(t)$, resp...

Answer: 2

Transfer function analysis. $G(s)$ in feedback system. Calculating gain for given parameters gives answer 2.

Source: Adapted from GATE 2024 EC Session 1 Q13

Q51. Network, Signals & Systems | 2 marks | 2NAT
A 50 lossless transmission line is terminated with a load Z_L of $(50 - j75) \Omega$. If the average incident power on the line is...

Answer: 0.48

50 Ω terminated with $Z_L = (50 - j75) \Omega$. Reflection coefficient and power calculation gives answer 0.48.

Source: Adapted from GATE 2025 EC Session 1 Q64

Q52. Network, Signals & Systems | 2 marks | 2MCQ
In the circuit shown below, the transistors M_1 and M_2 are biased in saturation. Their small signal transconductances are...

Answer: A

MOS transistors in saturation. Small signal analysis with given transconductances gives output matching option A.

Source: Adapted from GATE 2024 EC Session 1 Q51

Q53. Network, Signals & Systems | 2 marks | 2MCQ
Let (x, y, z, t) and $u(x, y, z, t)$ represent density and velocity, respectively, at a point (x, y, z) and time t . Assume is...

Answer: B

Continuity equation for charge and velocity. Analysis shows option B satisfies the given conditions.

Source: Adapted from GATE 2024 EC Session 1 Q25

Q54. Network, Signals & Systems | 2 marks | 2MCQ
Consider a continuous-time finite-energy signal $x(t)$ whose Fourier transform vanishes outside the frequency interval $[-f_m, f_m]$.

Answer: D

Multiplying the sampled signal by $\sin(\pi f_m t)$ gives the original baseband component only if the cutoff exceeds the signal bandwidth. This is equivalent to the condition given in option D). In that case the output of the filter is exactly the original signal $x(t)$.

Source: Adapted from GATE 2025 EC Session 1 Q38

Q55. Analog Circuits | 1 marks | 1MCQ
Common emitter has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q4

Q56. Network, Signals & Systems | 2 marks | 2MSQ
Consider a system represented by the block diagram shown below. Which of the following signal flow graphs represent(s) the system?

Answer: A,B

Block diagram to signal flow graph conversion. Multiple valid representations exist matching options A and B.

Source: Adapted from GATE 2025 EC Session 1 Q56

Q57. Communication Systems | 1 marks | 1MCQ
PCM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q4

Q58. Network, Signals & Systems | 2 marks | 2NAT
A full scale sinusoidal signal is applied to a 10-bit ADC. The fundamental signal component in the ADC output has a norm...

Answer: 2

10-bit ADC with full-scale sinusoidal input. SNR calculation gives fundamental component normalized value = 2.

Source: Adapted from GATE 2024 EC Session 1 Q46

Q59. Network, Signals & Systems | 2 marks | 2NAT
Consider the unity-negative-feedback system shown in Figure (i) below, where gain 0 . The root locus of this system is sh...

Answer: 3

Root locus of unity feedback system with gain $K > 0$. Analysis shows 3 branches/poles matching answer.

Source: Adapted from GATE 2025 EC Session 1 Q26

Q60. Network, Signals & Systems | 2 marks | 2NAT
A satellite attitude control system, as shown below, has a plant with transfer function $G(s) = 1$ cascaded with a compens...

Answer: 0.5

Satellite attitude control with compensator. Stability analysis gives damping ratio = 0.5.

Source: Adapted from GATE 2024 EC Session 1 Q38

Q61. Engineering Mathematics | 1 marks | 1MCQ
 $P(A \cap B) = ?$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q1

Q62. Network, Signals & Systems | 2 marks | 2MSQ
Which of the following can be used as an n-type dopant for silicon? Select the correct option(s).

Answer: A, D

Arsenic and Phosphorous are group V elements, providing extra electrons when doped into silicon, thus creating n-type conductivity. Boron and Gallium are group III elements and act as p-type dopants.

Source: Adapted from GATE 2025 EC Session 1 Q31

Q63. Network, Signals & Systems | 1 marks | 1MCQ
Sequence the following sentences (P, Q, R, S) in a coherent passage: P: Shifu's student exclaimed, "Why do you run since...

Answer: Q,S,P,R

Sequence the sentences coherently. Logical flow: Q (student question) !' S (master's answer) !' P (student action) !' R (resolution).

Source: Adapted from GATE 2024 EC Session 7 Q6

Q64. Network, Signals & Systems | 2 marks | 2MCQ
The relationship between any N -length sequence $x[n]$ and its corresponding N -point discrete Fourier transform $X[k]$ is defined...

Answer: A

DFT relationship: $x[n]$ and $X[k]$ are periodic with period N . Option A correctly describes this relationship.

Source: Adapted from GATE 2024 EC Session 1 Q59

Q65. Network, Signals & Systems | 2 marks | 2MSQ

Which of the following statements is/are TRUE with respect to ideal MOSFET- based DC-coupled single-stage amplifiers hav...

Answer: B, D

In an ideal MOSFET, the gate draws no DC current, giving the common source stage an infinite input resistance (B). The common drain (source follower) stage provides a non inverting voltage gain, so its input and output voltages are in phase (D). The common gate stage has low input resistance, not infinite (A false), and the common source stage inverts the signal, giving a 180° phase shift (C false).

Source: Adapted from GATE 2025 EC Session 1 Q30